

Dataset	Original Count	Undersampled Count	Split %
Training	42,910	7,551	60
Validation	14,304	2,517	20
Testing	14,304	2,518	20

Table 1: Comparison of dataset sizes before and after undersampling. The original dataset contained 42,910 training examples and 14,304 each for validation and testing. For model training, the data was undersampled to 7,551 training, 2,517 validation, and 2,518 testing examples, maintaining a 60/20/20 split.

Model/Classifier	Train Accuracy %	Test Accuracy %	F1 Score %
Logistic Regression (Baseline)	53.60	53.26	52.60
Random Forest	81.61	56.31	57.56
XGBoost	60.35	56.47	57.91
Transformer	55.94	52.70	46.04

Table 2: Model Performance on The Balanced Test Dataset